

*Teaching Notes 6A**

Modigliani-Miller (MM)

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Overview

- Last several lectures have focused on valuation and capital budgeting
- We will now move to the second part of the class and cover “MM”
 - Fundamental result in corporate finance
 - Will allow us to think about the “financing” decision
 - Sets the scene for the rest of the course

Objectives

- Understand the basic MM proposition
- Understand the underlying five assumptions that allow one to get the MM result

What will “MM” allow us to do?

- Recall....**Capital Budgeting**
 - Valuing projects to decide whether they are profitable
- **Capital Structure**
 - Issue 1: How to raise capital required to finance profitable projects?
 - Debt or equity
 - Issue 2: How a given capital structure affects value of the project/firm?
 - Already know debt tax shield increases value ($\Rightarrow$ use debt), but there are other effects to consider

The MM Proposition

Capital Structure: All About Debt and Equity

- **Debt: Market Value = D**
 - Debt holders receive interest & principal payments
 - e.g., corporate bonds, bank loans, revolving credit facilities, trade credit, etc.
- **Equity: Market Value = E**
 - Equity holders receive the residual value of the firm after debt holders paid (“residual claimants”)
 - Paid through dividends, share repurchases, and retained earnings
- **Value of a firm** is the market value of all its outstanding securities

$$V = D + E$$

Capital Structure: Relationship with Value

- Can firms create value by borrowing more?
 - Under what situations?

Value Creation: Increased Borrowing

Apple Bonds: 'Market Is Going to Be All Over It'

Jeff Cox | @JeffCoxCNBCcom

Published 10:55 AM ET Tue, 30 April 2013 | Updated 7:15 PM ET Tue, 30 April 2013



The tech giant Tuesday booked a **\$17 billion deal** that will come in six parts, with pricing of the floating- and fixed-rate debt combination near top-quality investment grade.

...

On its surface, the record-setting debt will be used to make good on a share buyback program.

...

Shares rose more than 3 percent in Tuesday trading.

Capital Structure: Relationship with Value

- Can firms create value by borrowing more?
 - Under what situations?
- Alternatively, can firms create value by repurchasing stocks, or from repurchasing debt?
 - Under what situations?

Value Creation: Buybacks

The Hot Stock: Anadarko Jumps 8.2%

By [Johanna Bennett](#) Sept. 21, 2017 4:30 p.m. ET

Anadarko Petroleum (APC) gushed to the top of the S&P 500 Thursday after the U.S. energy giant unveiled a **\$2.5 billion share repurchase plan**.

Anadarko's **share price rose more than 8.2% today** to close at \$48.48 a share.

Barron's, Sep 21 2017

Glenmark Pharma considers buyback of Singapore FCCBs; stock jumps 3%

ETMarkets.com | Sep 26, 2018, 10:18 AM IST

Shares of Glenmark Pharma **jumped over 3 per cent** in Wednesday's trade after the company said its operations committee would **consider opportunistic tenders or buybacks** of any part of its Singapore listed **foreign currency convertible bonds**.

Economic Times, Sep 26 2018

Capital Structure: Relationship with Value

- Can firms create value by borrowing more?
 - Under what situations?
- Alternatively, can firms create value by repurchasing stocks, or from repurchasing debt?
 - Under what situations?
- Can firms create value by paying dividend out of its earnings?
 - Under what situations?

Value Creation: Dividends

EBay to pay first-ever dividend

By **Ken Martin** | Published January 30, 2019 | **Ecommerce** | **FOXBusiness**

Ebay said Tuesday it would **pay a 14-cent dividend in March, its first ever.**

...

Ebay also added \$4 billion to its buyback program. Its authorization was \$3.2 billion on Dec. 31.

The dividend, which is set to be paid on March 20, is part of a broader capital-return plan the company announced.

Shares of eBay rose nearly 2.8 percent during after-hours trading.

Fox News, Jan 30 2019

Answers in Miller-Modigliani (MM) Proposition

- *MM Proposition:*
 - *“The total value of the securities issued by a firm is independent of the firm’s choice of capital structure. The firm’s value is determined by its real assets and growth opportunities, not by the types of securities it issues”*
- MM implies that capital structure does not matter for the value of the firm
- But, MM holds under certain assumptions
 - It is the precise nature of these assumptions that makes the MM proposition great

MM Assumptions

- MM requires the following five assumptions:
 - Perfect and complete capital markets
 - No Taxes
 - Bankruptcy is not costly
 - Capital Structure does not affect investment policy and cash-flows
 - Symmetric information
- Note: “only” these five assumptions

Proving MM Proposition

Proof of Modigliani-Miller (MM)

- Consider two firms, Lever (**L**) and Unlever (**U**)
 - Both have identical assets and cash flows
 - All cash flows paid out as either interest or dividends
- Capital structure of **L** and **U**:
 - **L** financed with debt requiring a payment of \$50 next year
 - **U** is all equity financed
- MM Proposition says values of **L** and **U** are the same
 - Is it?

Proof: No Arbitrage Condition (Law of One Price)

- Consider the following two portfolios:
 - Portfolio A: Buy 50% of U's equity
 - Portfolio B: Buy 50% of L's equity and 50% of L's debt

CF	Payoff A	Payoff B
10	5	$0 + 5 = 5$
50	25	$0 + 25 = 25$
100	50	$25 + 25 = 50$
200	100	$75 + 25 = 100$

- Value of 50% of U's Equity = Value of 50% of L's Equity + Value of 50% of L's Debt
 - Value of U = Value of L
- Homework: Show that you can prove this also by taking:
 - Portfolio C: Buy 50% of L's Equity
 - Portfolio D: Buy 50% of U's Equity and short 50% of L's Debt

Intuition behind MM

- Changing a firm's capital structure leaves the cash flows from assets unchanged
 - Investors can “undo” the change in capital structure by trading in the market on their own
- **Law of One Price/ No Arbitrage**
 - The two capital structures must have the same total value
 - Otherwise, investors can replicate them: short the expensive one and buy the cheap one
 - Incidentally, MM invented arbitrage pricing too!
- One Line Summary
 - Under some assumptions (“**a MM world**”), the size of the pie is determined solely by the value of firm's cash flows
 - Capital structure determines how it is sliced between owners of the firm but does not determine how large the pie is

MM Assumptions: Used in the Proof?

- Let's consider where we used the MM assumptions in the proof
 - Perfect and complete capital markets?
 - No Taxes?
 - Bankruptcy is not costly?
 - Capital Structure does not affect investment policy and cash-flows?
 - Symmetric information?

A Few Examples...

Example 1: Changing Capital Structure

- MM Inc. is financed solely by common stock. MM has 3m shares outstanding with a market price of \$12 per share. MM now announces that it intends to issue \$6m of debt, and use the proceeds to repurchase shares. [Note: We are in a “MM world”]
1. What is the market value ($E + D$) after this change in capital structure?
 2. How is the market price of the stock affected by the announcement?
 3. How many shares can MM Inc. repurchase with the \$6m million of new debt that we issue?

Example 1 (contd.): Changing Capital Structure

- Let P_{old} , P_{new} be old and new share prices
- Let S_{old} and S_{new} be old and new number of shares outstanding.
- $S_{new} = S_{old} - R$, where R is the number of shares repurchased.

Two Equations and Two Unknowns:

1. New Price: $P_{new} = (V - D)/(S_{old} - R) = (\$36 - \$6)/(3 - R)$
2. Number of Shares Repurchased: $R \times P_{new} = D = \$6$

Answer:

- Solve these equations, to get $P_{new} = \$12$ and $R = .5m$.
 - Repurchase $R = .5m$ shares, and new price $P_{new} = \$12$

Example 2: The P/E ratio

- “Eternity R Us” generates expected cash-flow of \$5 million per year in perpetuity
 - Debt rolled over every year with r_D of 4%
 - Market value of debt: $D = \$10$ million
 - Firm has 4 million shares, priced at \$10/share
 - Market value of equity: $E = \$40$ million
- Market value of the firm = $D + E = \$50$ million
- Note: We are in a “MM world”

Example 2 (contd.): The P/E ratio

- Questions:
 - Solve for r_A
 - Solve for Earnings per Share (EPS)
 - Solve for Price-Earnings ratio (P/E)
 - Solve for r_E

Example 2 (contd.): The P/E ratio

- r_A
FCFs are a Perpetuity $\rightarrow V = C/r_A$, so $r_A = 5/50 = 10\%$
- Earnings per share (EPS)
= (Revenue – Interest)/number of shares
= $(\$5 - (4\% * \$10)) / 4 = \$1.15/\text{share}$
- Price-Earnings ratio (P/E)
= $\$10 / \$1.15 = 8.70$
- r_E : Re-levering gives
 $r_E = 10\% + (10/40) * (10\% - 4\%) = 11.5\%$
 - Why is $r_E > r_A$?

Example 2 (contd.): The P/E ratio

- Firm takes additional capital structure decision
 - Announces it will issue new shares in order to buy back all outstanding debt
 - A “recap” decision
- Questions:
 - How many shares do we need to issue, and what happens to the share price?
 - What happens to the P/E ratio?
 - What happens to r_E ?

Example 2 (contd.): The P/E ratio

- Let S be the number of new shares issued, and let P_{new} be the new share price
- Without any debt, the value of the equity equals the value of the assets (\$50 million)

Two Equations and Two Unknowns:

- $P_{\text{new}} = \text{new value of equity} / \text{new total \# of shares}$
 $= \$50\text{m} / (4\text{m} + S)$
- $S \times P_{\text{new}} = \$10 \text{ million (buying back all debt)}$

Answer:

- Solving these gives
 - $S = 1$ million and $P_{\text{new}} = \$10$

Example 2 (contd.): The P/E ratio

- What happens to EPS?
 - Earnings per share decreases to $\$5/5 = \$1/\text{share}$
- What happens to the P/E ratio?
 - The P/E ratio goes up to $\$10/\$1=10$
- What happens to rate of return on equity?
 - $r_E = 10\% + (0/50) * (10\% - 4\%) = 10\%$
 - Why is now $r_E = r_A$?

Unlevering: MM Proposition II

r_E and r_A linked by MM Proposition

- Called second MM proposition:

$$r_E = r_A + \frac{D}{E}(r_A - r_D)$$

- **Cost of equity** = **business risk** + **financial risk**
 - Can express in terms of β 's too (see Lecture 4B)
- In Lecture 4B: used this formula to lever/unlever → “implicitly” assumed in “MM world”
 - In “non-MM” world, this formula may change (e.g., in presence of interest tax shield, Lecture 5 B)

The Real World is NOT MM World

- MM assumptions don't hold, and the proposition is not important because it is true
 - In reality, financing does matter!
- But, MM provides an important framework
 - Financing matters only when one or more of MM's assumptions is violated
 - Our task is to figure out “which one”
- Next few lectures will be devoted on how these violations can affect financing decisions

Takeaways: MM

- Value of a firm is the sum of market value of its debt and market value of its equity
 - $V = D + E$
- Under MM assumptions:
 - Capital structure does not matter for a firm's value
 - Capital structure decisions such as recapitalizing may change value of D and value of E (and EPS and P/E), but not total value of the firm (D+E)
- MM not important because right, but because gives framework to evaluate capital structure
 - Capital structure only matters when one or more of MM assumptions fail